

WARAGODA MUDALIGE RUCHIRA PERERA

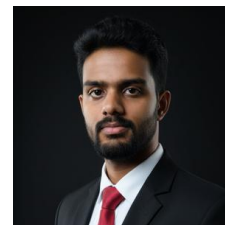
BSc (Applied Science) — Chemistry & Computer Science

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Location: Galle, Sri Lanka



PROFESSIONAL SUMMARY

Applied Science undergraduate with hands-on experience in advanced instrumentation (ICP-MS, GC-MS) and a strong interdisciplinary background in Chemistry and Computer Science. Published researcher with contributions to heavy-metal food safety analysis and AI-driven systems engineering. Passionate about combining laboratory science with technology to solve real-world problems, with a keen interest in analytical instrumentation and its applications across research and industry. Certified in Lean Project Management and proven leadership through academic and extracurricular achievements.

EDUCATION

B.Sc. in Applied Sciences

2022 – Present (Awaiting Graduation)

Faculty of Applied Sciences, Rajarata University of Sri Lanka, Mihinthale

- Major subjects: Chemistry and Computer Science

Computer Aided Drug Design (CADD)

July – August 2024

CINEC Campus, Malabe, Sri Lanka

- GPA: 3.30 / 4.00 (B+) — 5-week intensive programme

G.C.E. Advanced Level (2020/2021)

Siridhamma College, Galle

- Chemistry – B | Physics – C | Biology – C | General English – A

G.C.E. Ordinary Level (2017) – English Medium

Siridhamma College, Galle

- 9 Subjects — All A grades: Buddhism, Sinhala, English, Mathematics, Science, History, ICT, Tamil, English Literature

ANALYTICAL & TECHNICAL SKILLS

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| ► Mass Spectrometry: ICP-MS, GC-MS (hands-on) | ► Chromatography: GC |
| ► Spectroscopy: UV-Vis | ► AI / ML: SVM, Naïve Bayes, Edge-AI |
| ► Programming: Python, HTML, CSS, Node.js | ► Hardware / IoT: Raspberry Pi, servo-driven actuation |
| ► Computational Chemistry: Gaussian 09W, GaussView, HyperChem Professional, WinMOPAC, UCSF Chimera, PyRx (hands-on, CADD) | ► Productivity: Microsoft Office Suite (Word, Excel, PowerPoint) |
| ► Project Mgmt: Lean PM Yellow Belt certified | ► Data Analysis: Scientific data processing & reporting |

RESEARCH & PUBLICATIONS

Peer-Reviewed / Conference Publications:

[1] Varietal-Specific Assessment of Arsenic and Mercury Content and Associated Health Risks in Traditional Sri Lankan Rice — 30th International Forestry and Environment Symposium (IFES 2026), Dept. of Forestry & Environmental Science, University of Sri Jayewardenepura, 2026 [Published — Volume 30]

[2] Smart Eco Bin: An Autonomous Toxicity-aware Risk Scoring System for Intelligent Waste Segregation using Edge-AI and Servo-Driven Actuation — 12th Moratuwa Engineering Research Conference (MERCon 2026), Engineering Research Unit, University of Moratuwa | IEEE, 2026 [Submitted — Under Review]

[3] ResQNet: A Multi-Layer Hybrid AI Framework for Priority-Aware Disaster Communication over Decentralized Mesh Networks — International Conference on Technology Innovations for Crisis Management (ICTICM 2026), SLIIT (AIMS Research Center) & Massey University, New Zealand, 2026 [Accepted — Awaiting Presentation]

Industry Article:

[4] Learning to Learn in the Age of AI Marketing: How Prompt Engineering is Redefining Branding — TrendFluence — Department of Marketing Management, Rajarata University of Sri Lanka, 2025 [Published — Volume 01]

KEY RESEARCH PROJECTS

Heavy Metal Safety in Sri Lankan Rice Varieties 2025 – 2026

Faculty of Applied Sciences, RUSL — Final Year Chemistry Research

- Assessed arsenic and mercury content in traditional Sri Lankan rice using ICP-MS
- Conducted varietal-specific health risk assessments; findings published at an international symposium

Eugenol & Cinnamaldehyde Characterization in Sri Lankan Cinnamon [Ongoing]

Faculty of Applied Sciences, RUSL — Final Year Research Project (Chemistry)

- Performed methanol Soxhlet extraction on cinnamon leaf and cinnamon bark samples, followed by GC-MS profiling to identify and quantify eugenol and cinnamaldehyde as the principal bioactive constituents in their respective matrices
- Conducted a comprehensive DFT computational study of eugenol and cinnamaldehyde using Gaussian 09W and GaussView, encompassing ESP mapping, free energy, total energy, and UV absorption analysis; confirming strong antioxidant character, bioactive electronic profile'

Smart Eco-Bin — Edge-AI Waste Segregation System 2024 – 2026

Faculty of Applied Sciences, RUSL — Final Year Project (Computer Science)

- Developed an autonomous waste segregation prototype on Raspberry Pi 5, integrating a YOLOv8n edge-AI model trained across seven plastic resin and e-waste categories
- Implemented toxicity-aware risk scoring with servo-driven actuation for physical sorting

ResQNet — AI Disaster Communication Framework 2026

Independent Research

- Designed a multi-layer hybrid AI system using SVM and Naïve Bayes for priority-based message routing
- Targeted decentralized mesh network scenarios for post-disaster emergency communication

PROFESSIONAL CERTIFICATIONS

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| ▶ Lean Fundamentals Certified: LeanPM.org — Sep 2025 | ▶ LeanPM® Yellow Belt (LeanPM®-YB): Lean PM Foundation — Sep 2025 |
| ▶ Project Management Essentials: Management & Strategy Institute — Jun 2024 | ▶ Effective Leadership: HP LIFE Foundation — Jun 2024 |
| ▶ Introduction to Digital Business Skills: HP LIFE Foundation — Sep 2025 | ▶ Computer Aided Drug Design (CADD): CINEC Campus, Malabe — Aug 2024 |

LEADERSHIP & EXTRACURRICULAR ACHIEVEMENTS

Vice President — Chemistry Society 2025

Faculty of Applied Sciences, Rajarata University of Sri Lanka

- Led academic events and inter-faculty science activities.

Inter-University Quiz Competition — Varsity Battles 2025

Colombo Stock Exchange (CSE) — Island-wide Competition

- Member of the 2nd Runners-Up team in this prestigious national-level quiz competition

Senior Scout — District Commissioner's Medal Recipient

2014 – 2017

Siridhamma College Scout Troop, Galle

- Participated in international jamborees and Kanborees; led scouting teams across multiple activities
- Awarded the District Commissioner's Medal — highest district-level scout honour

Athletic & Sports Achievements

2019 – 2024

Siridhamma College & RUSL

- House Vice Captain at Annual Sports Meet 2019 — led house to Runners-up position
- 2nd Place — Road Race, Faculty Sports Meet, RUSL (2024)
- Member of Cricket and Elle teams — both achieving Runners-up positions

Volunteer Work — Zero Plastic Community Initiative

2025

Rajarata University of Sri Lanka

- Contributed to campus-wide environmental awareness and plastic waste reduction efforts

REFERENCES

Prof. Ranjith Edirisinghe

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